

Project Design Phase
Solution Architecture

Date	29 September 2026
Team ID	SWTID-2026-1542
Project Name	AnswerBot – AI-powered FAQ and Customer Support Automation
Maximum Marks	4 Marks

Solution Architecture:

AnswerBot is a layered, headless REST API platform that bridges the gap between the business problem (slow, manual FAQ creation and support) and the technology solution (AI-assisted, secure, searchable FAQ management). Its goals are to:

- Automate FAQ creation and answer drafting using Google Gemini AI.
- Describe the structure, components and data flow of the backend to project stakeholders.
- Define features, development phases and solution requirements.
- Provide specifications according to which the solution is defined, managed and delivered.

Architecture Components

- **Client Interface:** Postman / ThunderClient or any frontend sends HTTP requests (GET, POST, PUT, DELETE) to the API.
- **Express Server Gateway:** Receives requests on the configured port, parses JSON payloads, applies CORS and routes requests.
- **Authentication Middleware:** Extracts the JWT from request headers, verifies it and checks the user's role before allowing access to protected routes.
- **Controllers and Services (MVC):** Controllers validate requests and delegate work; services hold the logic for authentication, FAQ CRUD, category management and search.
- **Database Layer:** Mongoose schemas (User, FAQ, Category) enforce validation and translate operations into MongoDB queries.
- **AI Integration:** Google Gemini (@google/genai) generates questions, answers, categories and topic suggestions before they are saved.
- **Configuration:** Secrets such as the JWT secret, database URI and Gemini API key are kept in a .env file (dotenv).

Data Flow

- The client sends a request; Express parses it and passes it to the authentication middleware.
- Public routes (view/search FAQs) skip authentication; creator and admin routes require a valid JWT and the correct role.
- The controller calls the relevant service. For AI requests, the service sends a prompt to Gemini and returns the generated FAQ draft.

- Approved FAQs are stored in MongoDB through Mongoose, with the creator linked by the createdBy reference.
- A standardized JSON response (or centralized error message) is returned to the client.

Technology Stack

- **Runtime and framework:** Node.js (v24last), Express.js
- **Database:** MongoDB with Mongoose ODM
- **Security:** JWT and bcrypt
- **AI:** Google Gemini SDK
- **Tools:** VS Code, Postman/ThunderClient

Solution Architecture Diagram:

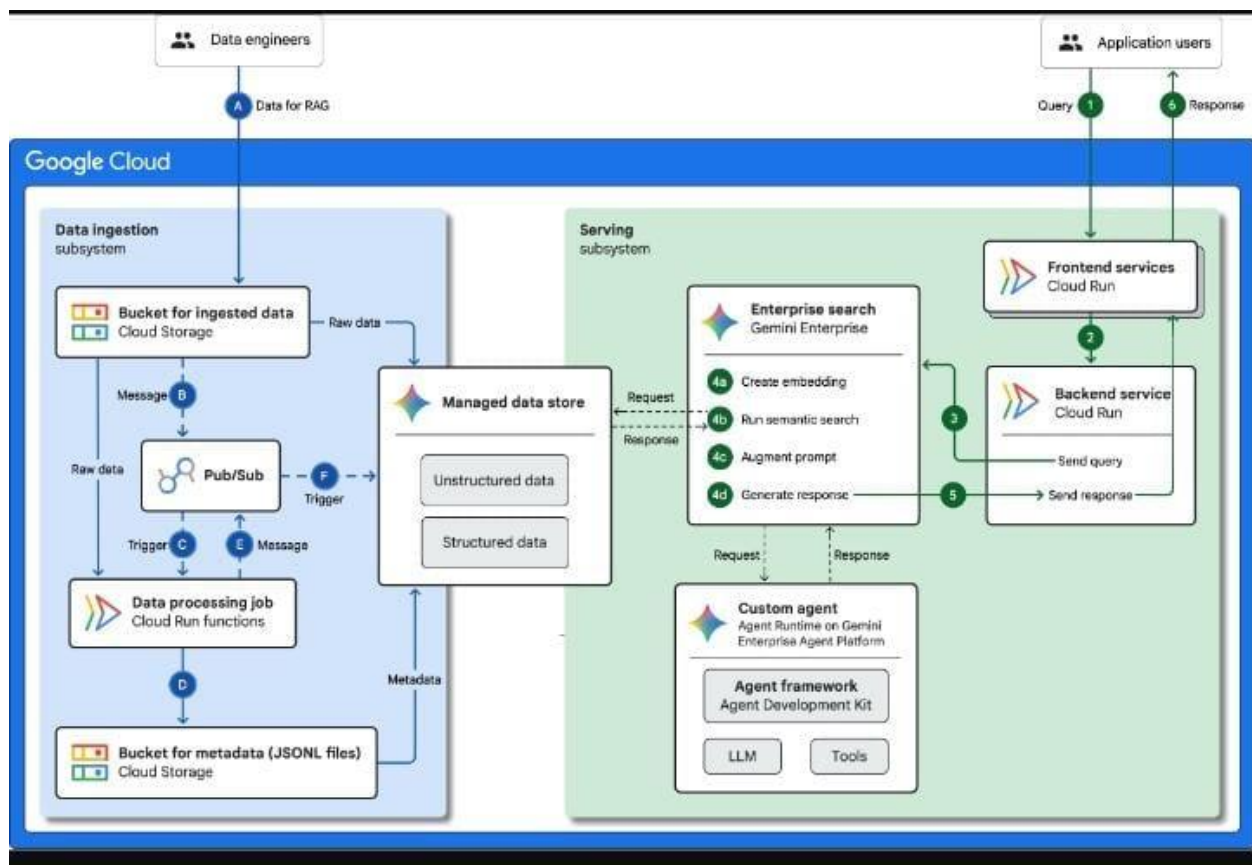


Figure 1: Architecture and data flow of the AnswerBot AI FAQ Assistant

References:

1. Google Cloud – Generative AI for customer support (reference architecture for AI-based FAQ and support answers):
<https://docs.cloud.google.com/architecture/genai-customer-support>
2. Google Cloud – Generative AI RAG reference architecture (retrieval, search and AI-generated answers from a knowledge base):

<https://docs.cloud.google.com/architecture/rag-genai-agentspace-vertexai>